

Learnable Stochastic Differential Equations for Movement Prediction

Author information to be added after approval

Draft — no empirical claims included

Abstract

Movement prediction for search and rescue must represent uncertainty rather than returning a single nominal path. This draft specifies a research paper around a learnable stochastic differential equation (SDE) implementation and a separate Lean formalization of selected probability identities. The executable baseline is a segment-level latent-regime underdamped Langevin model with exact-Gaussian, Euler–Maruyama, split-step, and common-random-number inference routes. The formal development checks algebraic statements for soft evidence and mixture weights. No private trajectory data, trained checkpoint, empirical metric, or performance conclusion is reported here. The experimental sections state a protocol to be completed only with approved, versioned aggregate evidence.

1 Introduction

Search planning benefits from a forecast distribution over plausible movement states at one or more future horizons. Such a distribution must retain the uncertainty induced by incomplete observations and heterogeneous movement behaviour. A stochastic differential equation provides a direct representation of state evolution while allowing the forecast to be sampled or summarized by moments.

This paper documents the present research artefacts without extending their claims. The software repository implements a typed Python framework for model fitting and inference. Its current executable baseline uses a latent regime that remains fixed inside a trajectory segment. The Lean repository formalizes a finite-dimensional, finite-measure algebraic core for soft evidence. Analytic existence, uniqueness, and Doob h -transform results are outside its stated formalization scope.

The paper will make three bounded contributions once the required evidence is available: an auditable implementation description, a precise account of the machine-checked theorem boundary, and a preregistered evaluation protocol. It does not claim that the current model improves any search-and-rescue outcome.

2 Problem formulation

Let $X_t = (p_t, v_t)^\top \in \mathbb{R}^2$ denote a position–velocity state for one coordinate of a movement segment. A contextual model may be written as

$$dX_t = b_\theta(t, X_t, c_t) dt + \sigma_\theta(t, X_t, c_t) dW_t, \tag{1}$$

where c_t contains approved contextual inputs and W_t is a Wiener process. The released repository does not include a registered neural SDE or a public context-data pipeline. Equation (1) therefore states the modelling interface, not an empirical model claim.

Given an observed initial state, the forecast task returns samples or moments of $X_{t+\tau}$ for requested horizons τ . Evaluation must use splits defined outside the source repository and report only approved aggregate statistics.

3 Implemented segment-level model

The current baseline is an affine underdamped Langevin SDE with a latent regime $r \in \{1, \dots, K\}$. The regime is supplied explicitly and is constant within each segment. Its dynamics are

$$dp_t = v_t dt, \tag{2}$$

$$dv_t = [-\Gamma_r v_t + (a_r - \kappa)p_t + c_r] dt + g_r dW_t. \tag{3}$$

Here Γ_r , a_r , c_r , and g_r are regime-specific parameters, while κ is a model parameter shared by regimes. The state transition is affine Gaussian, so the implementation provides an exact transition kernel when this model is selected.

The estimation code initializes segment labels from features derived from velocity and displacement variation. It then alternates segment posterior weights with weighted least-squares updates of a discrete transition, followed by a discrete-to-continuous conversion. This is an implementation description; convergence of the estimator on a target dataset remains an empirical question.

4 Inference and software boundary

The Python framework separates the SDE model, estimator, inference engine, evaluation rule, and artifact adapters. Exact-Gaussian sampling is available for models that provide an exact transition. Euler–Maruyama is an explicitly selected numerical route. The segment-level model also has a split-step route, and common random numbers can be used for shared-path, multi-horizon sampling. Unsupported combinations fail explicitly rather than silently selecting another method.

The repository’s synthetic workflow is deterministic and is intended for installation and regression tests. It is not evidence for real-data predictive accuracy. Neural-SDE and Fokker–Planck variants remain unregistered research skeletons, and condition-feature alignment has not fully moved into the declarative data-source layer.

5 Soft evidence and formal verification boundary

For binary soft evidence with prior π , likelihoods ℓ_1 and ℓ_0 , the formal development includes the posterior mixture weight

$$w = \frac{\ell_1 \pi}{\ell_1 \pi + \ell_0 (1 - \pi)}. \tag{4}$$

It checks the mixture identity, a likelihood-ratio form of Equation (4), a total-variation mixture bound, and properties of the logistic weight. The Lean development also covers a product identity for evidence-pattern weights and a nested-evidence concentration result under the assumptions encoded in its theorem signature.

These results are not a formal proof of SDE well-posedness or of a Doob h -transform for the target process. The manuscript must preserve this boundary: only declarations present in the Lean source and its axiom audit may be described as machine checked.

6 Data governance and reproducibility

The paper repository must not contain raw or transformed trajectories, coordinates, timestamps, identifiers, row-level predictions, non-public checkpoints, or archives containing such material. The current software contract expects a locally controlled data root with a main trajectory table and a split manifest. Its physical authoritative location and data steward are not established by this paper draft.

Every reported result should record a logical data version, split-manifest version, source revision of the Python and Lean repositories, random-seed policy, model configuration, and aggregate-only output. A controlled runner may mount the approved data root read-only. Public continuous integration should use only deterministic synthetic data.

7 Experimental protocol

This section is intentionally a protocol, not a result section. Before any table or conclusion is added, the following items must be fixed and reviewed:

1. the authorised dataset snapshot, split manifest, licence, and privacy classification;
2. forecast horizons, observation window, state construction, and missing-data handling;
3. baseline models and the exact inference configuration for each comparison;
4. aggregate scoring rules, including one explicit energy-score convention;
5. the number of seeds, uncertainty intervals, and the rule for handling failed runs;
6. a disclosure of known numerical limitations and any deviation from the preregistered plan.

Table 1 is a publication template. Its cells must remain blank until values are generated by an approved run and independently checked against the recorded configuration.

Table 1: Aggregate evaluation template. Values are intentionally absent.

Model	Horizon	Aggregate score	Run reference
Segment-level SDE	[to be specified]	[not yet available]	[to be specified]
Approved baseline	[to be specified]	[not yet available]	[to be specified]

8 Limitations and conclusion

The current artefacts establish an executable baseline, explicit numerical routing, deterministic synthetic tests, and a bounded formalization. They do not establish predictive performance on private trajectories, operational value for a search mission, or formal analytic properties beyond the Lean scope. The next revision should add only reviewed aggregate evidence and must update the limitations when the software or formal boundary changes.

Source-artifact note. Technical statements in this draft are traceable to the Python implementation repository and the Lean formalization repository. Bibliographic citations have deliberately not been invented; verified references belong in `references.bib` before submission.